

Practical 55 — C Programming Code: Login Validation System

Overview

This document presents the implementation of Practical 55 (Login Validation System Using Functions) in C programming language. The code modularizes login validation using a dedicated function `validate_Login()`, tracks failed attempts, and locks the account after 3 unsuccessful attempts.

Source Code

```
/*
 * Practical 55: Login Validation System Using Functions
 *
 * Requirements:
 * 1. Store predefined username and password.
 * 2. Accept login credentials from user.
 * 3. Validate credentials using a separate function validate_Login().
 * 4. Display appropriate success/failure messages.
 * 5. Lock account after 3 unsuccessful attempts.
 */

#include <stdio.h>
#include <string.h>

// Predefined stored credentials
#define STORED_USERNAME "admin"
#define STORED_PASSWORD "admin123"
#define MAX_ATTEMPTS 3

// Function Declaration
int validate_Login(char input_username[], char input_password[]);

int main() {
    char input_user[50];
    char input_pass[50];
    int attempts_left = MAX_ATTEMPTS;
    int is_logged_in = 0;
```

```

    printf("Applications of Programming\n");
    printf("-----\n\n");

    while (attempts_left > 0) {
        printf("Username : ");
        scanf("%s", input_user);

        printf("Password : ");
        scanf("%s", input_pass);

        // Call validation function
        if (validate_Login(input_user, input_pass)) {
            is_logged_in = 1;
            break;
        } else {
            attempts_left--;
            if (attempts_left > 0) {
                printf("\nInvalid credentials. %d attempt(s) remaining.\n\n",
attempts_left);
            }
        }
    }

    printf("\nSample Output\n");
    printf("-----\n");

    if (is_logged_in) {
        printf("Login Successful\n\n");
        printf("Welcome Administrator!\n");
    } else {
        printf("Account Locked!\n\n");
        printf("Maximum login attempts exceeded.\n");
    }

    return 0;
}

// Function Definition
int validate_Login(char input_username[], char input_password[]) {
    if (strcmp(input_username, STORED_USERNAME) == 0 &&
        strcmp(input_password, STORED_PASSWORD) == 0) {
        return 1; // Success
    }
    return 0; // Failure
}

```